Anabil's Travel Planner

README / Product & User Flow Documentation

A mobile-first AI-assisted travel planning web application designed to create personalized trip itineraries based on origin, destination, budget and trip duration.

Application Type	Mobile-first single-page travel planner
Primary Purpose	Generate personalized travel itineraries
AI Providers	Groq AI, Gemini AI, plus Offline mode
Budget Modes	Cheap, Medium, Premium
Trip Duration	2–10 days
Location	Optional current-location detection
Destination Search	Autocomplete / location suggestions
Booking Support	Hotel, train and bus shortcut links

1. What Is This Application?

Anabil's Travel Planner is a responsive travel-planning web app intended primarily for mobile users. The user enters where they are travelling from, chooses a destination, selects a budget level and chooses the number of days. The application then creates a structured travel plan containing destination information, a day-by-day itinerary, estimated budget guidance, local places and useful booking/map actions.

The application combines AI-generated planning with browser-based location services and destination suggestions. It is designed to reduce the effort normally required to search multiple websites before deciding how to travel and what to do.

2. Main Features

Feature	Description
AI Travel Planning	Creates a customized itinerary using the selected origin, destination, budget and number of days.
Three AI/Planning Modes	Groq AI for fast generation, Gemini AI as an alternative provider, and Offline mode when no API key is available.
Budget Selection	Cheap, Medium and Premium options influence the style and expected cost of the generated trip.
Flexible Duration	Users can choose from 2 through 10 days.
Current Location	The browser can request permission to access the phone's location and use it to auto-fill the From location.
Destination Autocomplete	The From and Destination fields provide suggestion boxes to help users select recognizable places.
Coordinates	The application can retain origin and destination latitude/longitude values for travel-distance and direction-related log
Travel Feasibility Check	The planner can warn users when the journey distance may consume too much of the available trip time.
Day-by-Day Itinerary	The generated result is organized into individual days with activities and destination imagery.
Budget Summary	The result includes a budget box to give the user an overall cost expectation.
Local Information	The result can present useful local places/services with Map, Number and Direction actions.
Booking Shortcuts	Hotel, train and bus buttons provide convenient next-step links.
Quick Destinations	Popular destinations such as Manali, Goa, Jaipur, Rishikesh, Leh, Paris, Bali, Dubai, Bangkok and Tokyo can be sel
Mobile-first UI	The interface uses cards, large touch-friendly buttons, compact forms and responsive layouts.
Developer / Support Footer	The application includes developer contact, Instagram and a UPI-based coffee-support option.

3. User Interaction Flow

The complete user journey can be understood as the following sequence:

Step 1 — Open the App

The user opens the travel planner. The home screen shows the app title, provider selection, location/origin fields, destination field, budget options and trip duration.

Step 2 — Select Planning Provider

The user chooses Groq AI, Gemini AI or Offline. For an AI provider, the user can paste an API key and save it. Offline mode is intended for situations where an API key is not being used.

Step 3 — Enter Origin

The user can type a city/place into From. Alternatively, the user taps “Use my current location” and grants browser location permission. The app then uses the detected coordinates to help identify the origin.

Step 4 — Enter Destination

The user types the destination. The autocomplete area can display matching place suggestions, making it easier to select the intended location.

Step 5 — Choose Budget

The user selects Cheap, Medium or Premium. Medium is the default selection.

Step 6 — Choose Trip Length

The user selects the number of days, from 2 to 10.

Step 7 — Start Planning

The user taps “Plan My Trip”. The application validates the input and starts the itinerary-generation process.

Step 8 — Feasibility Check

When coordinates/distance information is available, the app can estimate whether the travel distance is practical for the selected number of days. If the journey appears too time-consuming, a warning can be shown.

Step 9 — User Decision

If a feasibility warning appears, the user can change the plan or continue anyway. This prevents an unrealistic itinerary from being accepted without the user's awareness.

Step 10 — Generate Itinerary

The selected planning provider generates the travel plan. A loading indicator is shown while the application works.

Step 11 — Review Results

The user receives a destination card, trip metadata, day-by-day activities, budget guidance, tips and local information.

Step 12 — Take Action

The user can open map/direction links, call useful local numbers, and use hotel/train/bus shortcuts to continue planning or booking.

4. AI Provider Flow

The provider selector gives the application three planning paths. The exact implementation of the API calls is handled by the JavaScript portion of the application.

Mode	Purpose	Typical User Flow
Groq AI	Fast AI itinerary generation.	Select Groq → enter/save key → enter trip details → Plan My Trip.
Gemini AI	Alternative AI provider.	Select Gemini → enter/save Gemini key → enter trip details → Plan My Trip.
Offline	Use the planner without an AI API key.	Select Offline → enter trip details → Plan My Trip.

5. Location & Distance Flow

Location is optional. If the user chooses to use the current location, the browser asks for permission. The application stores latitude, longitude and accuracy information and can use the coordinates as the origin. The destination can also have coordinates when selected through the destination-location workflow.

The current code defines a planning rule of approximately 400 km of travel per full travel day and a travel-day ratio limit of 50% of the total trip. This is used as a practical feasibility heuristic rather than a guarantee of real-world travel time.

6. Feasibility Warning Flow

• User enters From, Destination and number of days.

• The app evaluates available distance information.

• If the journey appears to consume an excessive share of the trip, a yellow feasibility warning can be displayed.

• The warning explains the concern and provides two choices: change the plan or continue.

• The user remains in control; the warning does not automatically cancel the trip.

7. Results Page / Generated Trip

Destination overview: Destination name, travel metadata and a destination image.

Daily itinerary: Separate Day 1, Day 2, etc. sections with activities.

Budget: A highlighted budget estimate/summary based on the selected planning style.

Tips: Travel advice and practical notes.

Local contacts/places: Useful local entries with Map, Number and Direction actions where available.

Booking actions: Hotel, Train and Bus shortcut buttons.

8. Screen / Component Structure

Area	What the user sees / does
Header	Application title and travel quote.
Provider section	AI provider selector, API key field, save button and provider help/status.
From field	Origin entry with autocomplete.
Location button	Requests current phone/browser location and can auto-fill From.
Destination field	Destination entry with autocomplete.
Budget selector	Cheap / Medium / Premium.
Duration selector	2–10 day trip duration.
Plan button	Starts validation and itinerary generation.
Quick select	One-tap selection of popular destinations.
Loading state	Spinner and progress message while planning.
Results area	Generated itinerary and travel information.
Footer	Developer information, phone, Instagram and support/UPI link.

9. Error & Loading Experience

The UI includes a dedicated loading state and an error box. This allows the user to understand that the application is processing the request rather than appearing frozen. Errors can be presented in a visible red warning area instead of failing silently.

10. Design & UX Principles

• Mobile-first layout optimized for phone screens.

• Large touch-friendly controls and rounded cards.

• Sage/green visual theme representing travel, nature and calm.

• Clear separation between trip input, planning and generated results.

• Progressive disclosure: users only see detailed results after requesting a plan.

• Quick-select destinations reduce typing for common trips.

• Location is permission-based and optional.

• Warnings are actionable rather than simply blocking the user.

11. Example User Journey

Example: A user in Kolkata wants to visit Manali for 5 days with a medium budget.

→ Open Anabil's Travel Planner.

→ Keep Groq AI selected and provide the required API key.

→ Tap “Use my current location” and allow location access, or type Kolkata manually.

→ Enter/select Manali as the destination.

→ Select Medium budget.

→ Select 5 Days.

→ Tap “Plan My Trip”.

→ Review any feasibility warning and either modify or continue.

→ Wait while the itinerary is generated.

→ Review the day-by-day Manali plan, budget and tips.

→ Use local Map/Direction actions and booking shortcuts when needed.

12. Important Security / Deployment Note

The supplied source currently contains a Groq API key directly in client-side JavaScript. Any API key embedded in a public HTML/JavaScript file should be considered exposed because visitors can inspect the page source or browser requests. For a public deployment, the recommended architecture is to keep the API key on a server-side backend or use a secure serverless proxy, with rate limits and usage controls.

Location data should also be handled transparently. The app should request browser permission only when needed and explain why location is being used. If the application stores or transmits coordinates, a privacy notice should clearly explain retention and use.

13. Future Enhancement Ideas

• Real-time train, bus and flight availability.

• Live hotel pricing and availability.

• Weather forecast integration for the selected travel dates.

• Map-based route visualization.

• Automatic travel-time calculation using a routing service.

• Save/share itinerary as PDF.

• Trip history and saved destinations.

• User accounts and cloud-saved trips.

• Emergency contacts and nearby police/hospital information.

• More accurate budget calculations using current prices.

• Backend API-key protection and request-rate limiting.

Document prepared from the supplied Anabil's Travel Planner HTML/CSS/JavaScript source.